RETAIN 9/29

Question of the day - Dogs or cats?

Goals:

- **Determine an estimate for the size of the water tank**
- **Learn more about NOLOP!**

Agenda:

- **Split up into two groups - one will go to NOLOP for a short tour first, and then they will switch**
- **For the tank:**
  - **Break down the engineering design process**
  - **Define the problem and brainstorm solutions**
  - **Make a list of considerations (size, materials, structure, design...)**

RETAIN 10/6

Question of the day - What should replace the JCC's Starbucks?

Goals

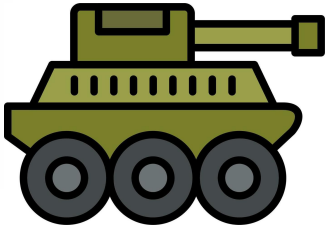
- Decide and write out equations for water tank design
- Determine potential tank locations and the number of tanks needed
- Make an outline for October goals

Agenda

- Discuss data with the MAPS team
  - How large are certain community buildings?
  - Based on their data and rainfall averages, determine the quantities of water collected from those buildings
- Review and discuss equations and design considerations from the last meeting
- Determine potential tank locations and how many we would need
- Outline goals for the month

RETAIN 10/13





Question of the Day - Is water wet?

Goals

- Become water storage tank and plumbing experts

Agenda

- Why are military tanks called tanks?
- Figure out how many tanks we want to have and the best positions to have them
  - Locations need to have altitude measurements so we can determine flow directions
- Tank considerations
  - Everything that could go wrong with a tank
  - Worst thing that could go wrong wins
- Let's talk piping!
  - Learn more about the old PVC pipe water system that they had from the old water source

RETAIN 10/20

Question of the Day - would you rather visit the past or future?

Goals

- Discuss meeting this past week with Professor Limbrunner
- Make a plan for the rest of the semester

Agenda

- Find a better daily precipitation estimate and rainfall patterns
  - Create a mass/balance spreadsheet + simulation based on previous
    - Volume flow
    - Edit volume/depth of tank to observe changes in storage volume over time
  - Consider size of the roof
- Tank considerations
  - Water demand and use
    - We can have two tanks, each for different uses (one that undergoes more filtration)
  - Overflow
  - Withdraw
- Figure out and begin mapping the ideal locations for the tank
  - Locations need to have altitude measurements so we can determine flow directions

RETAIN 10/20

Question of the Day - what has been your best halloween costume ever?

Goals

- Get everyone to join the slack channel
- Make a plan for the rest of the semester
    - Final CAD model for tank (OnShape)
- Figure out mass balance simulation
- Determine size of tank

Agenda

- Discuss tech lead meeting discussion
    - Goal for semester = 2 tanks for one big pavilion
- Create a mass/balance spreadsheet + simulation based on previous

RETAIN 11/3

Question of the Day - if you could be any animal, which one would you be and why?

Goals

- Make a plan for the rest of the semester
  - Final CAD model for tank (OnShape)
- Employ the mass balance simulation to determine the size of the tank

Agenda

- Input rainfall data into the simulation and simulate with various storage sizes



RETAIN 11/10

Question of the Day - what's the first thing you're gonna do when you get home later?

Goals

- Make a plan for the rest of the semester
  - Final CAD model for tank (OnShape)
- Employ the mass balance simulation to determine the size of the tank
- Look at the data analysis for rainfall over the last 25 years

Agenda

- Input rainfall data into the simulation and simulate with various storage sizes to determine necessary storage size
- Consider strategies to account for long dry spells

RETAIN 11/17

Question of the Day - If you were to open any sort of restaurant, bakery, or food shop? What food would you sell?

Goals

- Make a plan for the rest of the semester
  - Final CAD model for tank (OnShape)
- Employ the mass balance simulation to determine the size of the tank
- Look at the data analysis for rainfall over the last 25 years

Agenda

- Split into groups for the different components of the tank
- Start designing our model in OnShape!